Hydrogen Garden — Report Outline & Fact Table

Nov 9, 2026 · @issa

How to use this doc

This is the plan for our 4-page report (the main deliverable). Every number or claim in the report, site and video must come from the **Master fact table** at the bottom.

1. **No table row, no claim.** If a fact is not in the table, it does not go in the report, the site or the video.
2. **Two-person rule.** Anyone can add a row. The Evidence & Critic (Person 3) checks it against the source and fills in "Checked by".
3. **Tag everything.** Each row is either **Evidence** (a real source says it) or **Assumption** (our choice). Fictional organisms are marked **Fiction**.
4. **Log every change.** When we change a decision, add one line to the decision log on page 4: what, why, who.
5. **Plain words.** Write for a high schooler. Define every science word the first time it appears.

Deadline check: the brief says "submit by 10 Nov, 14:00" on pages 2 and 4, but page 3 says the 24-hour window ends "9 Nov 14:00". Email the organizer today to confirm which one is right.

Rules from the brief: at most 4 pages, text no smaller than 10 pt, including visuals and references. The file is named `TEAMNAME_Challenge-A_Report.pdf` with no spaces in the team name.

Page budget

The report fits exactly 4 pages with about 1,500 words in total. The main visual takes up half of page 1.

Page	Report sections (from the brief)	Rubric points it mainly earns	Owner	Words (about)
1	1 Team · 2 Problem and users · 3 Solution · 5 Main visual	Problem 20, Clarity 10	P5 (team, visual), P4 (problem), P2 (solution)	350 + visual
2	4 Supporting science + evidence table	Science 20	P1	450 + table
3	4 (cont.) Food web and energy · Rejected alternative · 6 Limits and validation	Science 20, Creativity 20, Feasibility 20	P2 (food web), P1 (alternative), P3 (limits, validation)	450 + small diagram
4	7 Contributions · Decision log · AI disclosure · References	Teamwork 10, Science (sources)	P5 (contributions, log), P3 (AI, references)	300 + tables

Layout rules for P5: body text 10–11 pt, captions at least 10 pt. Put the site link and its QR code in the page-1 header, plus the video link if we make one. References in short form (author, year, journal, DOI) to save space.

Page 1 — Who we are, the problem, our idea

Section 1 · Team identification (P5, 2 lines). Team name, Challenge A: Life Without Sunlight, and all 6 member names with their roles.

Section 2 · Problem and intended users (P4, about 120 words).

- Problem: high-school students find it hard to imagine a food web with no sunlight at the bottom of it. Every food chain they learn starts with a plant.
- Users: students aged 14–18 and the science teachers who teach them. Scientists are not our audience.
- Narrow focus: one habitat zone (a vent field), one energy pathway (hydrogen), and a 4-level food web.
- Starter sentence: *"Every food chain in a school textbook starts with sunlight. We set out to show a high schooler, in under five minutes, how a whole ecosystem could run without it."*

- Add one line from the user test once it has run, for example how many of the testers could name the energy source before and after using the site. **Report only real results.**

Section 3 · Proposed solution and how it works (P2 and P6, about 200 words).

- The idea: the Hydrogen Garden, a fictional vent field 50 km down in the ocean of an icy moon we invented, modelled on Saturn's moon Enceladus.
- How the parts fit: **(1)** the report explains the science, **(2)** the interactive dive lets a student scroll from the ice surface down to the vents and tap each label for its fact and source, and **(3)** the video walks through it in 3 minutes.
- Key sentence: *"Rock reacts with water and releases hydrogen. Microbes turn that hydrogen into food. Three levels of animals live off those microbes."*
- P6 adds 2 lines on the creature mode: at each level the student plays that creature and follows where its energy goes.
- Put the site link + QR code here.

Section 5 · Main visual (P5, half the page). A cross-section from the ice down to the vents, with energy-flow arrows: hydrogen → methane-makers → scrapers / fan-feeders → Shadowfin, with the heat lost at each step. Label at least 4 conditions directly on the drawing (depth, pressure, temperature, chemicals, darkness). Caption: *"Energy flow in the Hydrogen Garden. Green = evidence, amber = our assumption."*

Page 2 — The science (Section 4, owner P1)

4a · The habitat zone and its conditions (about 150 words). The brief needs at least 4 conditions, so we state 6:

Condition	Our value	Evidence or assumption
Depth	50 km below the surface (20 km of ice + a 30 km ocean)	Based on Enceladus (NASA; Thomas 2016). The exact numbers are our choice
Pressure	about 56 bar at the seafloor, which equals ~550 m deep on Earth	Calculated: density × gravity × depth
Temperature	vent fluid 40–90 °C; surrounding water about 8 °C	Fluids: Lost City (Kelley 2005). Water at 8 °C: **assumption**

Condition	Our value	Evidence or assumption
Chemicals	hydrogen (H_2), carbon dioxide (CO_2), methane; salty, very alkaline water	Waite 2017; Postberg 2009; Kelley 2005
Light	none. Sunlight at Saturn is ~1% of Earth's and cannot get through 20 km of ice	NASA Saturn facts + the inverse-square law
Food from above	none (no "marine snow")	NOAA marine snow, applied by us

4b · The energy source (about 150 words). Seawater reacts with rock rich in olivine (serpentinisation) and releases hydrogen gas. Methane-making microbes use it:

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CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O
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- Why it lasts: tidal heating from Saturn keeps the rocky core warm. Models suggest hydrothermal activity could last tens of millions to billions of years (Choblet 2017). This answers guiding question 1.
- Where we've seen it: Lost City on Earth (Kelley 2005), and hydrogen in Enceladus' plume (Waite 2017). This answers guiding question 2.

4c · Design choices linked to evidence (the table the brief asks for).

#	Design choice	Real-world evidence	Source
1	Energy comes from hydrogen made when water reacts with rock	Lost City's vents are driven by this reaction; Cassini found H_2 in Enceladus' plume	Kelley 2005; Waite 2017
2	Methane-making microbes form the base of the food web	Methane-cycling microbes live at Lost City; an Earth methane-maker grew under Enceladus-like conditions	Kelley 2005; Taubner 2018
3	The habitat is a vent field in a dark, salty, global ocean under ice	Enceladus has a global ocean, salts in its ice grains, and silica grains that point to hot water reacting with rock	Thomas 2016; Postberg 2009; Hsu 2015

#	Design choice	Real-world evidence	Source
4	The energy supply lasts long enough for an ecosystem	Tidal heating can power long-lived hydrothermal activity	Choblet 2017
5	The food web is short and the top predator is small	~10% of energy passes between levels on average	Pauly & Christensen 1995
6	Life clusters at the vents and the open water is nearly empty	Earth's deep sea relies on food sinking from the sunlit surface, which our moon has none of	NOAA (marine snow)

Guiding question 3 (how conditions shape organisms): one sentence each. No light means no eyes; low energy means slow, small animals; microbes are fine at vent temperatures, since one methane-maker grows at up to 122 °C (Takai 2008).

Page 3 — Food web, the rejected alternative, limits and testing

4d · Food web and energy loss (P2, about 150 words + a small diagram by P5).

1. **Methane-makers** (Level 1): microbe mats on the chimneys that turn H_2 + CO_2 into food.
2. **Scrapers** (Level 2): snail-like grazers that scrape off the mats. *Fiction*, based on the snails and limpets at Earth's vents (NOAA poster).
3. **Fan-feeders** (Level 3): catch drifting microbes, scraps and scraper waste. *Fiction*, based on filter feeders at cold seeps (NOAA poster).
4. **Shadowfin** (Level 4): an eyeless, hand-sized predator. *Fiction.*

- Energy loss: start with 1,000 "energy coins" at Level 1. At ~10% per step (Pauly & Christensen 1995) that leaves 100, then 10, then 1 coin for Shadowfin. Say clearly that the coins are illustrative. This answers guiding question 4.
- If the vent stops: the mats starve first, then the grazers and filter feeders, then Shadowfin. Life continues only where new vents open (NOAA: vents are short-lived). This answers guiding question 5.

Creativity · The alternative we rejected (P1, about 100 words). This is worth up to 20 points, so don't skip it.

- Alternative: copy Earth's famous black-smoker vents, with giant tubeworms living on hydrogen-sulfide-eating bacteria.
- Why we rejected it: NOAA writes that reaction as $CO_2 + 4H_2S + O_2 \rightarrow CH_2O + 4S + 3H_2O$. It **needs oxygen**, and Earth's deep-sea oxygen comes from photosynthesis at the sunlit surface. A moon with no sunlight has no such oxygen supply. Hydrogen + CO_2 needs no oxygen, and both were found in Enceladus' plume (Waite 2017).
- Starter sentence: *"Earth's 'sunless' vent ecosystems secretly still depend on the Sun. Ours does not."*
- Optional second comparison (1 line, our reasoning only): why a scroll-down dive instead of a static poster. Scrolling makes 50 km of depth feel real.

Section 6 · Feasibility, limitations and validation (P3, about 200 words).

What is realistic: every chemical step has been observed on Earth or measured at Enceladus. Whether life actually exists there is unknown: *habitable does not mean inhabited.*

Limitations (the brief asks for at least 2):

- Our organisms are invented from Earth examples, and real alien life could be very different.
- The ~10% rule is an Earth average that varies a lot. Making methane also yields less energy than breathing oxygen, so the real food web could be even smaller.
- Our gauges use simplified physics: constant gravity, a straight-line temperature change through the ice, and a made-up light fade rate.
- Our depth, temperature and chimney numbers are borrowed from Enceladus and Lost City. They are not measurements of our fictional moon.

Validation plan, written as what / who / how / what counts as success:

Test	Who	How	Success means
Can the microbes live there?	A microbiology lab	Grow an Earth methane-maker at our garden's pressure, temperature, alkalinity and gas mix, as Taubner et al. (2018) did for Enceladus-like conditions	The microbes grow and produce methane

Test	Who	How	Success means
Is the chemistry there?	A future spacecraft that flies through the plume	Measure hydrogen, CO_2 and methane in the jets	All three are present. If there is less hydrogen than non-living chemistry predicts, that is a **hint, not proof**, that something is using it
Does our concept teach?	Us, with 3–5 high-school students	Ask a 3-question quiz, let them use the site for 5 minutes, then ask the quiz again	Our target: most testers can name the energy source and explain why the top predator is small afterwards. **Report the real results, whatever they are**

Page 4 — Teamwork, AI disclosure, references (Section 7)

7a · Contributions (P5). Be specific: name each person's actual output, never "helped with research". Fill in the names.

Member	Role	What they made	Report part they wrote
(name)	P1 · Science Lead	Habitat conditions, energy-source research, the black-smoker comparison	4a–4c, the rejected alternative
(name)	P2 · Ecology & Story	The 4-level food web, the energy-coin model, the "vent stops" scenario, the creature texts, the video script	3, 4d
(name)	P3 · Evidence & Critic	Checked every row of the fact table, the limitations, the validation plan	6, 7c, references

Member	Role	What they made	Report part they wrote
(name)	P4 · Video	The 3-minute video, the user test with students	2
(name)	P5 · Artist & Editor	All art layers, the main cross-section, the 4-page layout	1, 5, 7a, 7b
(name)	P6 · Web	The interactive dive, the creature mode, hosting, the QR code	3 (site part)

7b · Decision log (shared decisions and changes, newest first). This is what earns the full 10 teamwork points. Keep adding rows.

What we decided	Why	Decided by
Every member owns a report section, not only a video or site job	Points come from the report; the video and site earn none on their own	Team (fill in)
Dropped the predator's "glow sense" and changed it to sensing chemistry, warmth and movement	Vents at 40–90 °C are too cool to give off visible light	Team (fill in)
Cited the NOAA poster and fact sheet instead of the NOAA lesson link	The lesson link in the brief returned a 404 error	Team (fill in)
Built a scroll-down dive (one of 6 site ideas)	Scrolling makes the 50 km depth and the changing conditions feel real	Team (fill in)
Hydrogen + CO_2 microbes instead of sulfide vents	Sulfide chemistry needs oxygen made by photosynthesis	Team (fill in)
Chose Challenge A over B	(fill in the team's reason)	Team (fill in)

7c · AI-use disclosure (P3). Draft it, then edit until every word is true:

> *We used Claude (Anthropic) to brainstorm ideas, plan our work split, write code for the interactive site and suggest wording. We checked every source exists by opening its DOI or web page, and checked each number against its source. All final decisions and edits were made by our team. Reused work: none.* (List any images or tools we reused.)

7d · References (P3). Use short form; the brief needs at least 4, with at least 2 from an agency or peer-reviewed. We have 9 peer-reviewed and 6 agency sources.

1. Waite et al. (2017). *Science* 356, 155–159. doi:10.1126/science.aai8703
2. Hsu et al. (2015). *Nature* 519, 207–210. doi:10.1038/nature14262
3. Kelley et al. (2005). *Science* 307, 1428–1434. doi:10.1126/science.1102556
4. Taubner et al. (2018). *Nature Communications* 9, 748. doi:10.1038/s41467-018-02876-y
5. Takai et al. (2008). *PNAS* 105, 10949–10954. doi:10.1073/pnas.0712334105
6. Pauly & Christensen (1995). *Nature* 374, 255–257. doi:10.1038/374255a0
7. Postberg et al. (2009). *Nature* 459, 1098–1101. doi:10.1038/nature08046
8. Choblet et al. (2017). *Nature Astronomy* 1, 841–847. doi:10.1038/s41550-017-0289-8
9. Thomas et al. (2016). *Icarus* 264, 37–47. doi:10.1016/j.icarus.2015.08.037
10. NASA: Enceladus
11. NASA: Saturn facts
12. NASA (2017): Missions provide new insights into ocean worlds
13. NOAA: What is marine snow?
14. NOAA Ocean Exploration: Chemosynthesis fact sheet
15. NOAA Ocean Exploration: Life Beyond Light poster

Master fact table

There are 32 facts so far. 17 were pre-checked by AI against their source, and P3 must re-read each one and set it to **Team checked**. 6 still need a team check, 1 still needs a source, and 8 are our own choices. The site's `js/content.js` uses these same numbers, so change a number in both places.

#	Claim	Value	Source	Type	Status
1	Saturn's distance from the Sun	9.5 × Earth's (9.5 AU)	NASA Saturn facts	Evidence	AI pre-checked

#	Claim	Value	Source	Type	Status
2	Sunlight reaching Saturn	~1.1% of Earth's ($1 \div 9.5^2$)	Inverse-square law, using row 1	Calculated	AI pre-checked
3	Enceladus surface temperature	about −201 °C	NASA Enceladus	Evidence	AI pre-checked
4	Enceladus ice shell thickness	20–25 km on average; 1–5 km at the south pole	NASA Enceladus	Evidence	AI pre-checked
5	Our moon's ice shell	20 km	Our choice, based on row 4	Assumption	Our choice
6	A global ocean under the ice	yes	Thomas et al. 2016	Evidence	AI pre-checked
7	Our moon's ocean depth	30 km	Our choice (Enceladus: tens of km)	Assumption	Our choice
8	The ocean is salty	sodium salts in the plume's ice grains	Postberg et al. 2009	Evidence	AI pre-checked
9	Hot water reacts with rock	silica grains that form in water above ~90 °C	Hsu et al. 2015; NASA Enceladus	Evidence	AI pre-checked
10	Hydrogen gas in the plume	H_2 detected by Cassini	Waite et al. 2017	Evidence	AI pre-checked
11	CO_2 in the plume	present	Waite et al. 2017	Evidence	Needs check
12	The heat supply lasts	tens of millions to billions of years	Choblet et al. 2017	Evidence	Needs check
13	Serpentinisation makes H_2 at Lost City	yes	Kelley et al. 2005	Evidence	Needs check
14	Lost City chimneys and fluids	up to ~60 m tall; fluids ~40–90 °C; high pH	Kelley et al. 2005	Evidence	Needs check
15	Methane-cycling microbes at Lost City	yes	Kelley et al. 2005	Evidence	Needs check

#	Claim	Value	Source	Type	Status
16	The methane-maker reaction	$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$	Standard chemistry; Taubner et al. 2018	Evidence	AI pre-checked
17	A methane-maker grew under Enceladus-like conditions	yes, in a lab	Taubner et al. 2018	Evidence	AI pre-checked
18	The hottest methane-maker known	grows at up to 122 °C	Takai et al. 2008	Evidence	AI pre-checked
19	Earth's sulfide vent reaction needs oxygen	$CO_2 + 4H_2S + O_2 \rightarrow CH_2O + 4S + 3H_2O$	NOAA fact sheet	Evidence	AI pre-checked
20	Tubeworms rely on sulfide bacteria; snails and limpets live on vents	yes	NOAA poster	Evidence	AI pre-checked
21	Filter feeders live at cold seeps	sponges, corals, anemones	NOAA poster	Evidence	AI pre-checked
22	Vents are short-lived	yes	NOAA poster	Evidence	AI pre-checked
23	Deep-sea food webs depend on marine snow	yes	NOAA marine snow	Evidence	AI pre-checked
24	Energy passed between food-web levels	~10% on average	Pauly & Christensen 1995	Evidence	Needs check
25	Enceladus surface gravity	0.113 m/s²	**none yet**: find a NASA or peer-reviewed page	Evidence	Needs source
26	Pressure at our seafloor	~56 bar (the same as ~551 m deep on Earth)	Density × gravity × depth, using rows 5, 7, 25 and 27	Calculated	AI pre-checked
27	Ice and salty-water density	920 and 1,030 kg/m³	Standard textbook values	Assumption	Our choice
28	Water temperature in the garden	~8 °C	None: warmed a little by the vents	Assumption	Our choice

#	Claim	Value	Source	Type	Status
29	Temperature through the ice	a straight line from −201 °C to 0 °C	A simplification	Assumption	Our choice
30	How fast light fades in ice	halves about every metre (illustrative)	None: made up for the gauge	Assumption	Our choice
31	Scrapers, Fan-feeders, Shadowfin and the moon's name	invented	Based on the Earth analogs in rows 20–21	Fiction	Our choice
32	Energy coins	1,000 → 100 → 10 → 1	Illustrative, from row 24	Calculated	Our choice

Open items before submitting

- [ ] P3: find a cited source for Enceladus' gravity (row 25), or remove the pressure number from the report and the site
- [ ] P3: check rows 13–15 against Kelley et al. 2005 (the full paper is paywalled, so try a library or the abstract)
- [ ] P3: check rows 11, 12 and 24 against their papers' abstracts
- [ ] P3: re-read the 17 rows marked AI pre-checked and set each one to Team checked
- [ ] Team: choose the moon's name and the team name (no spaces), then update `js/content.js` and page 1
- [ ] P4: run the user test and write down the real results for sections 2 and 6
- [ ] P5: confirm the PDF is exactly 4 pages with no text under 10 pt
- [ ] Someone: email the organizer to confirm the deadline (9 Nov or 10 Nov, 14:00)
- [ ] P3: finalise the AI disclosure so that every word of it is true